SANGJUN LEE

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EDUCATION

University of California, Berkeley **MAY 2027**
Bachelor of Science in Electrical Engineering and Computer Science *Military Leave of Absence: Fall 2023 - Spring 2025*
Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Computer Security, Designing Information Devices and Systems, Discrete Mathematics, Probability, Multivariable Calculus, Structure and Interpretation of Computer Programs, Data Science
Teaching & Leadership: Computer Architecture and Machine Structures (uGSI & AI for CS 61C), The Foundations of Data Science (uGSI for Data 8), Mobile Developers at Berkeley (Contract & DevOps Team), Computer Science Mentors (Junior Mentor for CS 61C)

SKILLS

Languages: Python, Java, C, Kotlin, JavaScript, TypeScript, HTML, CSS, Swift, Go, SQL, RISC-V Assembly
Frameworks & Tools: Spring Boot, Linux, React, Node.js, Express.js, FastAPI, Docker, Git, PyTorch, NumPy, Pandas, PostgreSQL, CUDA

EXPERIENCE

Samsung — JUNE 2025 - AUG 2025
Software Engineer Intern — *South Korea*
- Engineered a novel GUI automation agent to overcome the API and security limitations of traditional AI agents, successfully automating complex workflows of over 15 actions and cutting manual task completion time by approximately 80%
- Implemented a FastAPI backend integrating multiple MLLM models with PyAutoGUI to translate user prompt into on-screen actions
- Built a React frontend allowing users to submit prompts and displaying each stage of model processing and actions in real time
- Boosted MLLM accuracy (Llama 4, UGround) by 30% through systematic prompt engineering and testing of over 20 distinct variations

Republic of Korea Air Force Headquarters — OCT 2023 - MAY 2025
Backend Developer — *South Korea*
- Served as a backend developer for an automated runway crack detection system, reducing inspection and labeling time by over 95%
- Implemented a Spring Boot backend with Oracle Database, managing 100,000+ runway slabs and permissions for 5,000+ users
- Led the full migration of the backend system from Java to Kotlin to boost long-term maintainability and enhance code safety
- Deployed an inference pipeline that uses YOLOv8 & OpenCV to label 4000+ images per hour, increasing detection accuracy by 35%

Machine Learning Engineer — *South Korea*
- Developed the FastAPI backend for a real-time transcription service, leveraging multithreading and concurrency to handle simultaneous model requests efficiently, reducing manual transcription time from weeks to minutes for 200+ daily users
- Increased transcription accuracy by 50% by training and fine-tuning STT and diarization models using PyTorch on large-scale datasets
- Boosted deployment time by 200%, using Docker for containerization and Kubernetes for orchestration and automated deployments
- Awarded the Prime Minister Award as part of the team that substantially reduced soldiers' administrative workload through AI

CS 61C Course Staff — JUNE 2023 - AUG 2023
Undergraduate Student Instructor — *Berkeley, CA*
- Instructed for a computer architecture class of 200+ students, leading to a 90% increase in student participation in discussion sections
- Created weekly discussion worksheets on core concepts such as C, RISC-V, hardware, and optimization (SIMD, OpenMP, OpenMPI)

AlectoAI — JUNE 2023 - AUG 2023
Software Engineer Intern — *Berkeley, CA*
- Developed a scalable Swift backend for an AI facial recognition app with features like comments and likes to boost user engagement
- Managed and constructed the Firebase database schema to ensure efficient data storage and retrieval while maintaining performance

PROJECTS

Rook
- Built a book recommendation platform with React, JavaScript, and Express.js, integrating Google Books API for search and metadata
- Implemented AI-driven recommendations by applying prompt engineering to enhance personalization and user engagement
- Designed secure authentication and authorization with JWT, integrating middleware for protected routes and user data integrity

The Locker Room
- Built a sports social media app using JavaScript, React, Node.js, and Supabase, enabling users to discuss about games and sports news
- Implemented the frontend Post screen, allowing real-time chat, comments, and likes on sports posts with text, images, and videos
- Developed the backend architecture and managed the database to efficiently store sports posts, user data, and real-time chat

Java Version Control System
- Developed a version-control system that mimics Git's core functionalities such as staging, committing, branching, and merging
- Incorporated serialization and hashing (SHA-1) to efficiently save files and commit trees, ensuring persistence across versions